

1 kovacicODE

1.1 Introduction to kovacicODE

1.2 Definitions for kovacicODE

`kovacicODE (expr, y, x)` [Function]

`kovacicODE` is an implementation of the Kovacic algorithm for finding the solution of second order linear ordinary differential equations (ODEs) with Liouvillian solutions. If the ODE has a Liouvillian solution, `kovacicODE` finds and returns the solution. If the ODE does not have a Liouvillian solution, `kovacicODE` returns `false`.

`load("kovacicODE")` loads this function.

Example:

Example 1 from "On Liouvillian Solutions of Linear Differential Equations" by F. Unger (1992), *Applicable Algebra in Engineering, Communication and Computing*, volume 2, issue 3, pp 171–193. `DEBUGFLAG` controls debugging output in `kovacicODE`; we'll set it to 0 to suppress debugging output in this example.

```
(%i1) load (kovacicODE) $
(%i2) eq : 'diff(y,x,2) - 'diff(y,x,1)/x+x^2*y/(x^4+2*x^2+1) = 0;
```

$$\frac{d^2 y}{dx^2} - \frac{dy}{dx} + \frac{x^2 y}{x^4 + 2x^2 + 1} = 0$$

```
(%o2)
(%i3) DEBUGFLAG : 0 $
(%i4) kovacicODE (eq, y, x);
```

$$[y = \%k2 \sqrt{x^2 + 1} \log(x^2 + 1) + \%k1 \sqrt{x^2 + 1}]$$

Appendix A Function and variable index

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